

EVENTFLOW AI

Infosys _Agentic AI for Smart Event Management Operations

MILESTONE 2 REPORT

Venue & Speaker Operations

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Program: Infosys Springboard Internship 7.0 — AI Theme

Mentor: Springboard Mentor 56

Repository / Stack: Python · Flask · SQLAlchemy · SQLite · Chart.js

1. Milestone 2 Objective

Milestone 2 extends EventFlow AI from registration and attendee management into intelligent venue and speaker operations. The objective is to automate venue allocation, manage speaker profiles and availability, schedule sessions without conflicts, optimize venue utilization, and generate session and operational analytics for improved event planning.

1.1 Milestone 2 Requirements

Milestone 2 Requirement	Implementation
Build Venue Agent	AI-based venue optimization and recommendation
Build Speaker Agent	Speaker profile, expertise and availability management
Develop Venue Optimization Workflows	Capacity, facility, utilization and scoring-based venue selection
Implement Speaker Scheduling	Speaker-session assignment with conflict detection
Generate Session Analytics	Analytics Agent with session, speaker, venue and attendance metrics

2. Venue Management Module

The Venue Management module allows organizers to register venues and lets the AI Venue Optimization Agent recommend the most suitable venue for an event based on capacity, facilities, availability and space efficiency.

2.1 Features

- Add and manage venues
- Store venue capacity
- Store facilities
- Track availability
- Match venue capacity with event requirements
- Match required facilities
- Calculate venue utilization
- Calculate unused capacity
- Rank suitable venues
- Recommend optimal venue

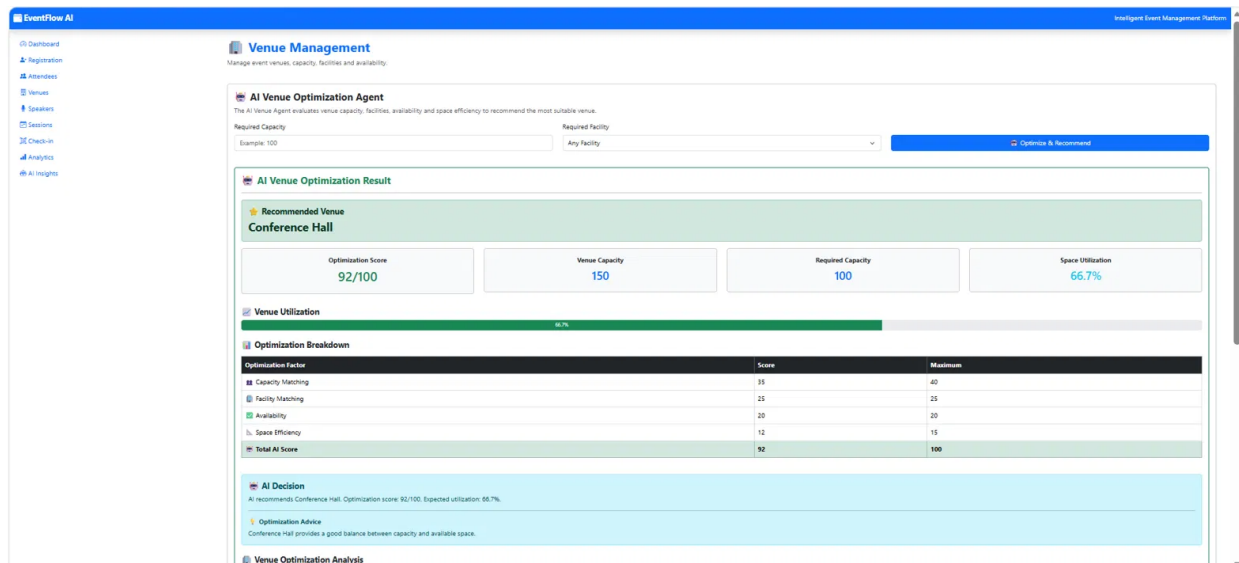


Figure 1: Venue Management — AI Venue Optimization Agent and recommendation result

2.2 AI Venue Recommendation

Given a required capacity and an optional required facility, the Venue Agent evaluates every available venue and returns the recommended venue together with an optimization score and a full breakdown of how that score was calculated (capacity matching, facility matching, availability and space efficiency).

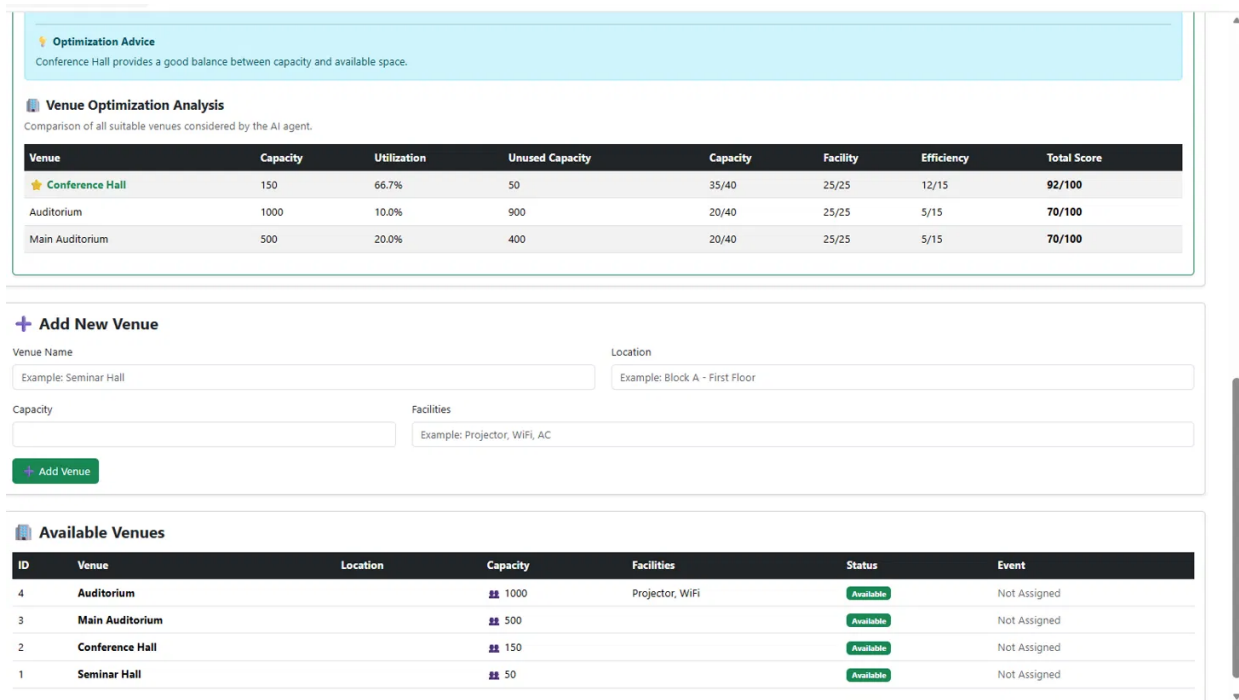
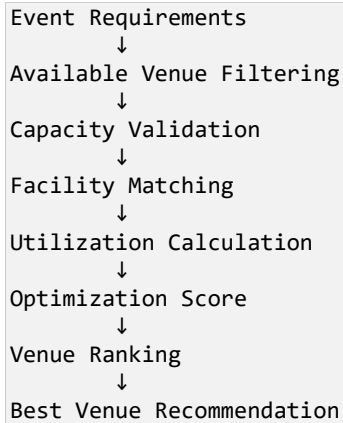


Figure 2: Venue Optimization Analysis — ranked comparison of all suitable venues with utilization and score

The table above shows that the system does not simply pick a venue at random — it filters out venues that cannot accommodate the required capacity, then scores and ranks the remaining venues (Conference Hall, Auditorium, Main Auditorium) on capacity, facility match, availability and efficiency before recommending the best option.

3. Venue Optimization Agent

Implemented in services/venue_agent.py, the Venue Agent is the core AI component behind venue allocation.



The Venue Agent evaluates available venues according to attendee capacity, required facilities and space utilization. Venues that cannot accommodate the required number of attendees are eliminated. Remaining venues are scored and ranked, allowing the system to recommend the most suitable venue while minimizing unused space, and to suggest a smaller venue when utilization would otherwise be low.

4. Speaker Management Module

The Speaker Management module maintains a database of speakers along with their expertise, experience and availability, and supports assigning speakers to events and sessions.

4.1 Features

- Speaker registration
- Speaker profile management
- Expertise tracking
- Experience tracking
- Availability tracking
- Assigned event
- Assigned session

 **Speaker Management**

Manage event speakers and their expertise.

Add Speaker

Speaker Name

Enter speaker name

Email

speaker@gmail.com

Expertise

AI, Machine Learning, Cloud

Experience (Years)

Availability

Available

Add Speaker

 **AI Speaker Recommendation**

Find the best speaker for a session based on expertise and experience.

Session Topic

Example: Artificial Intelligence

Required Experience

0

Find Best Speaker

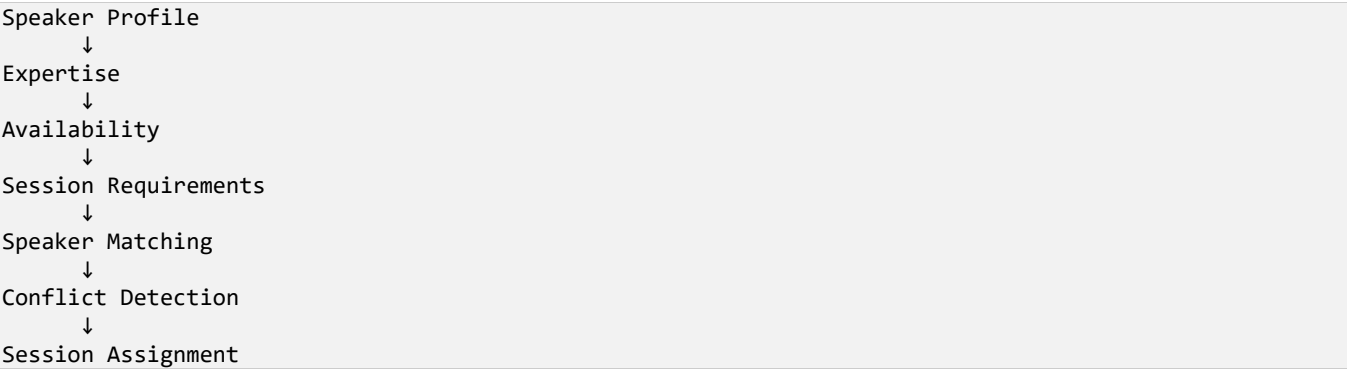
Registered Speakers

ID	Name	Email	Expertise	Experience	Availability
5	Siya	siya@gmail.com	Full Stack	4 years	Available
4	Advay	advay@gmail.com	Data Science	3 years	Available
3	Riya	riya@gmail.com	Cloud Computing	10 years	Available
2	Arjun	arjun@gmail.com	Machine Learning	8 years	Available
1	Akshitha	akshitha@gmail.com	AI	5 years	Available

Figure 3: Speaker Management Module — speaker registration, expertise and availability

5. Speaker Agent

Implemented in services/speaker_agent.py, the Speaker Agent matches speakers to a session topic based on expertise and experience, and returns a ranked list of AI-recommended speakers with a match score.



AI Speaker Recommendation
Find the best speaker for a session based on expertise and experience.

Session Topic: Required Experience:

[Find Best Speaker](#)

AI Recommendations
Best available speakers for: Full Stack

Siya Full Stack 4 years experience Expertise Match: 60/60 Experience: 30/30 Availability: 10/10	100% AI Match Score
Advay Data Science 3 years experience Expertise Match: 10/60 Experience: 30/30 Availability: 10/10	50% AI Match Score
Akshitha AI 5 years experience Expertise Match: 10/60 Experience: 30/30 Availability: 10/10	50% AI Match Score
Arjun Machine Learning 8 years experience Expertise Match: 10/60 Experience: 30/30	50% AI Match Score

Figure 4: AI Speaker Recommendation — expertise and experience-based match scoring for a session topic

As shown above, for the session topic “Full Stack”, the agent ranks all speakers by an AI match score built from expertise match, experience and availability, helping organizers pick the most suitable speaker rather than assigning one manually.

The speaker management module maintains speaker availability and supports speaker-session assignment and scheduling. Notification automation (e.g. automated speaker reminders) can be extended in a future iteration.

6. Intelligent Session Scheduling

The scheduling module assigns speakers and venues to sessions while validating date and time conflicts. Before a schedule is approved, the system checks whether the selected speaker or venue is already assigned to another session during the same time period.


Session Scheduling

Create sessions and let the AI Scheduling Agent detect speaker and venue conflicts.

Create Session

Session Title

Event
Select Event

Date

Start Time

End Time

Speaker
Select Speaker

Venue
Select Venue

Validate & Schedule


Schedule Approved

Schedule is valid. No speaker or venue conflicts detected.


Scheduled Sessions

ID	Session	Event	Date	Time	Speaker	Venue	Status
5	Cloud Computing	Cloud Summit	2007-08-15	10:00:00 - 12:00:00	Riya	Main Auditorium	Scheduled
1	AI	AI Workshop	2026-08-10	10:00:00 - 12:00:00	Akshitha	Conference Hall	Scheduled
3	Generative AI Workshop	AI Workshop	2026-08-11	10:00:00 - 12:00:00	Arjun	Main Auditorium	Scheduled
4	ML	Hackathon	2026-08-11	12:00:00 - 13:00:00	Arjun	Conference Hall	Scheduled
2	Generative AI Workshop	AI Workshop	2026-08-12	10:00:00 - 12:00:00	Akshitha	Main Auditorium	Scheduled

Figure 5: Session Scheduling — Schedule Approved, no speaker or venue conflicts detected

6.1 Speaker Conflict Detection

The scheduler cross-checks the selected speaker's existing session times before approving a new schedule. For example, if Speaker Akshitha already has a session from 10:00–12:00 and a new session is created for the same speaker in the same window, the system raises a speaker conflict instead of allowing a double-booking.

6.2 Venue Conflict Detection

The scheduling system also validates venue availability for each session. If two sessions attempt to use the same venue during overlapping time intervals, the system identifies the conflict and prevents the schedule from being approved.

Session Scheduling

Create sessions and let the AI Scheduling Agent detect speaker and venue conflicts.

Session Title

Example: Introduction to Generative AI

Event

Select Event

Date

dd-mm-yyyy

Start Time

--:--

End Time

--:--

Speaker

Select Speaker

Venue

Select Venue

Validate & Schedule

Scheduling Conflict

The session cannot be scheduled because:

Main Auditorium is already occupied by 'Generative AI Workshop' during this time.

Scheduled Sessions

ID	Session	Event	Date	Time	Speaker	Venue	Status
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1	AI	AI Workshop	2026-08-10	10:00:00 - 12:00:00	Akshitha	Conference Hall	Scheduled
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4	ML	Hackathon	2026-08-11	12:00:00 - 13:00:00	Arjun	Conference Hall	Scheduled
2	Generative AI Workshop	AI Workshop	2026-08-12	10:00:00 - 12:00:00	Akshitha	Main Auditorium	Scheduled

Figure 6: Venue Conflict Detection — scheduling blocked because Main Auditorium is already booked for that time slot

This directly demonstrates that the scheduler validates real constraints rather than simply storing whatever session data is submitted, satisfying the requirement to optimize room utilization and avoid scheduling conflicts.

7. Session & Operational Analytics

Implemented in services/analytics_agent.py, the Analytics Agent goes beyond the Milestone 1 category/event charts to analyze attendance, event popularity, venues, speakers and sessions, and to generate rule-based recommendations for organizers.

7.1 Metrics Generated

- Total attendees, checked-in attendees, pending attendees and attendance rate
- Event popularity (most popular event by registrations)
- Attendee category distribution (Students, Professionals, Speakers)
- Total venues, available venues and venue availability percentage
- Total speakers and available speakers
- Total sessions, scheduled sessions and schedule rate

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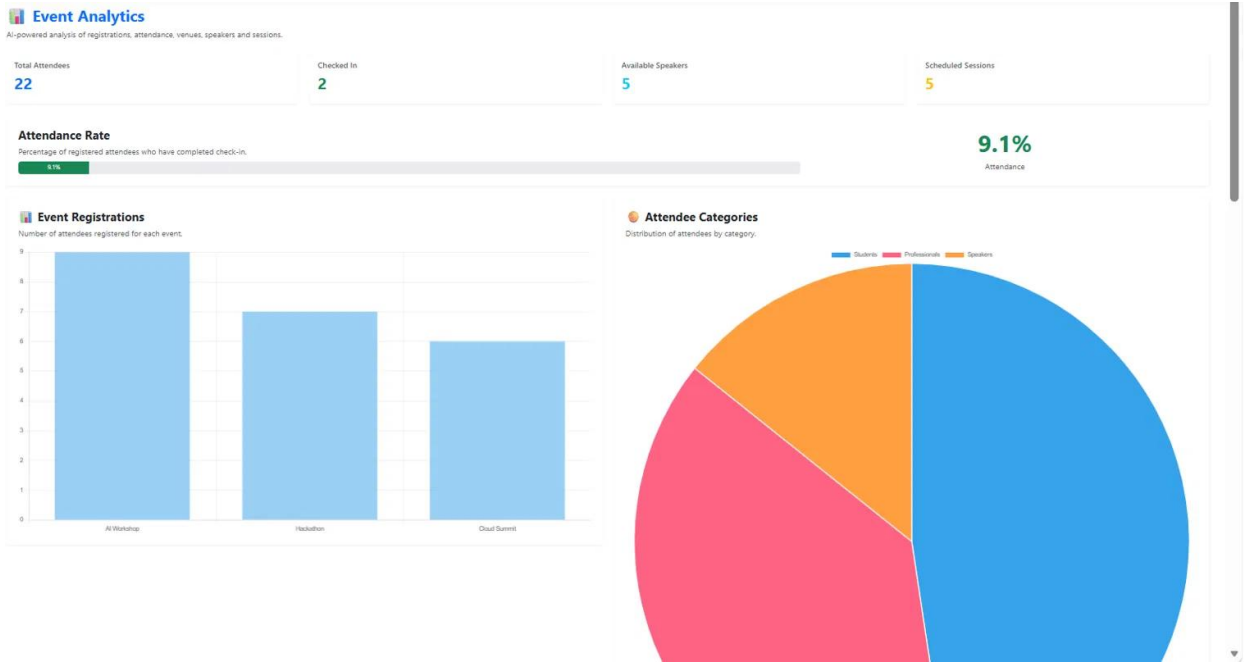


Figure 7: Event Analytics — attendance rate, event registrations and attendee categories

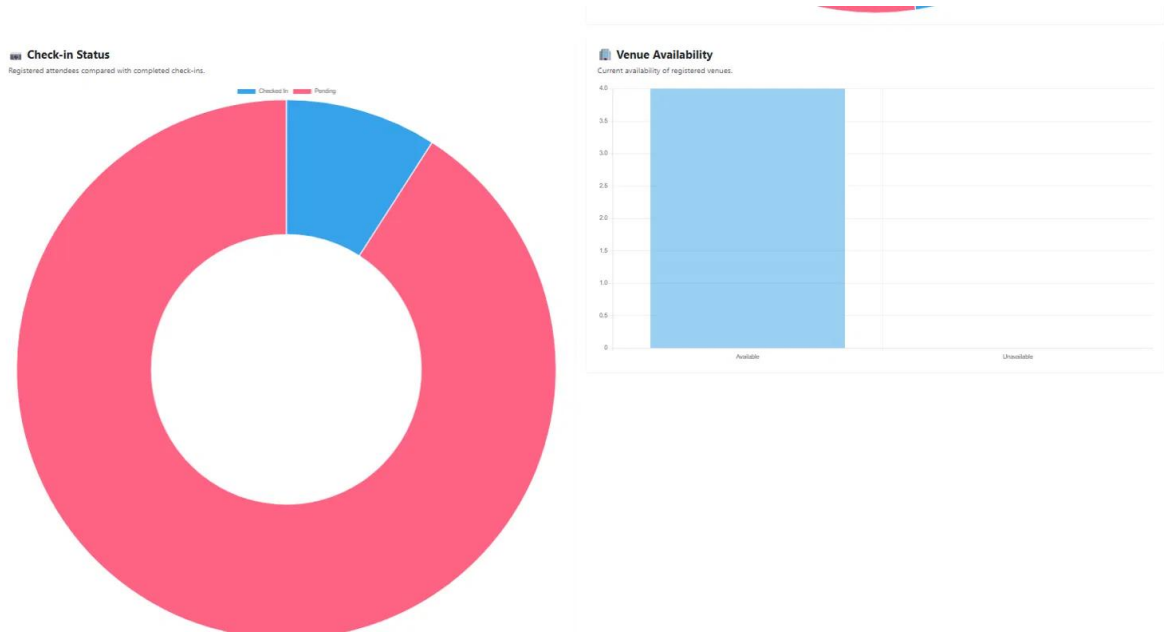


Figure 8: Check-in Status and Venue Availability analytics

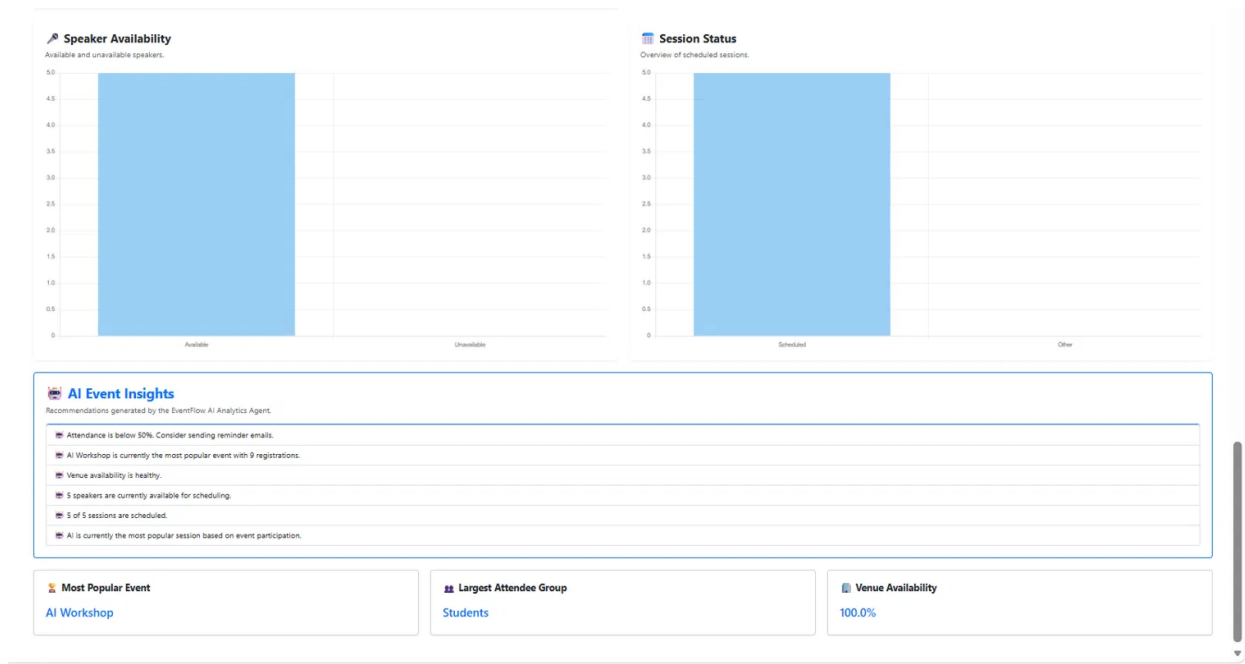


Figure 9: Speaker Availability and Session Status analytics, with AI-generated event insights

The AI Event Insights panel produces short, rule-based recommendations directly from the underlying data — for example flagging that attendance is below 50% and reminder emails should be considered, or identifying the most popular event and session based on participation. This satisfies the requirement to generate session analytics.

7.2 AI Insights Dashboard — Venue & Speaker Intelligence

The AI Insights Dashboard was extended for Milestone 2 to include venue and speaker intelligence alongside the existing event insight panel: per-venue utilization, speaker availability with session counts, and session popularity ranking.

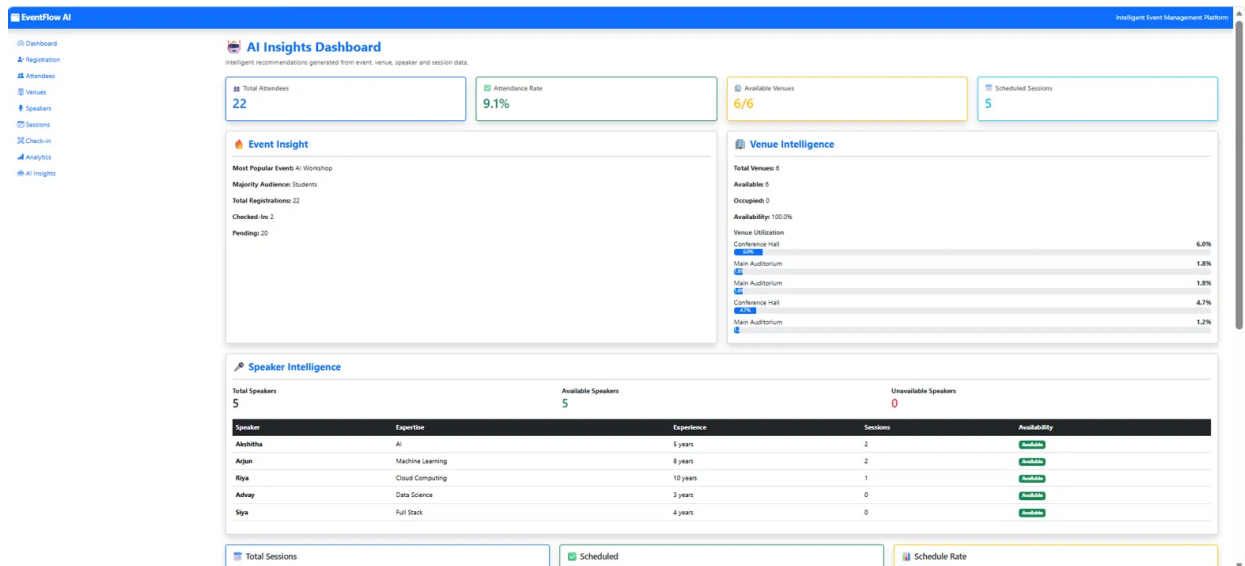


Figure 10: AI Insights Dashboard — Event Insight, Venue Intelligence and Speaker Intelligence

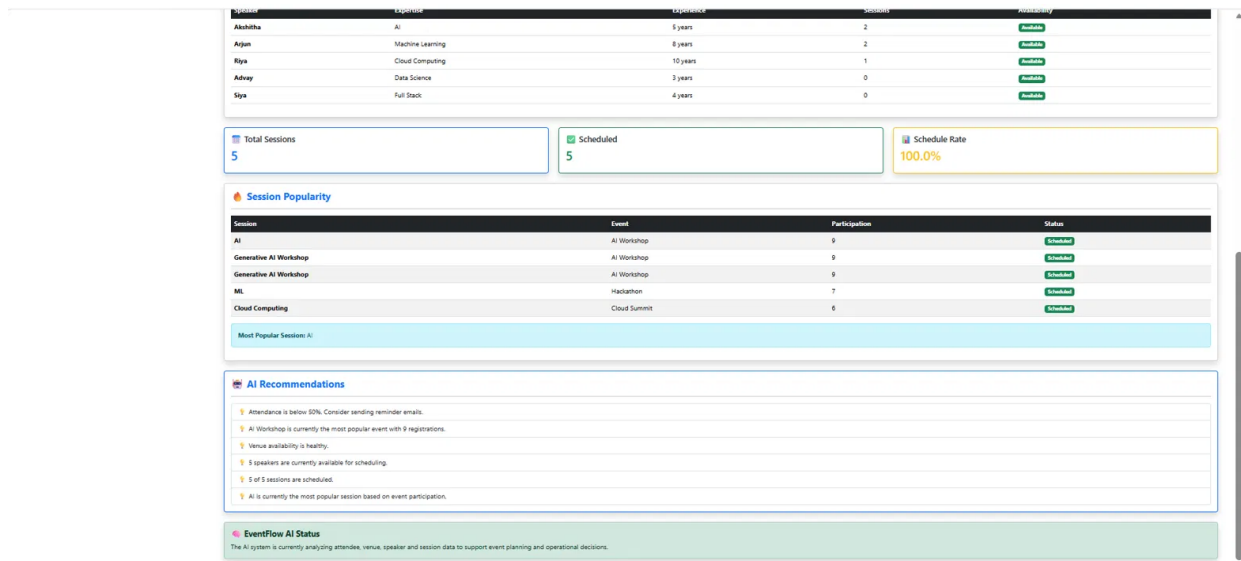
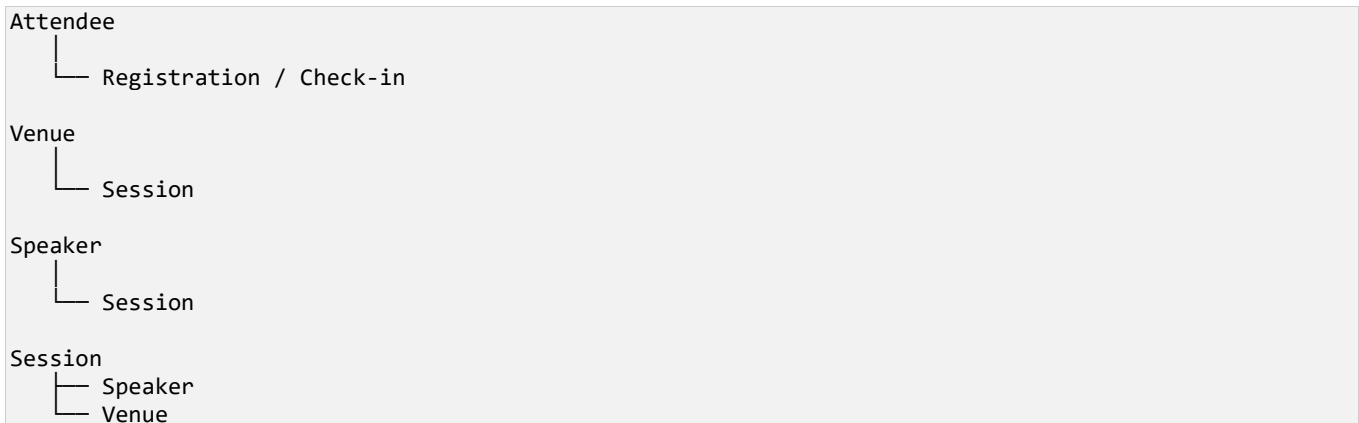


Figure 11: AI Insights Dashboard (continued) — speaker table, session popularity and AI recommendations

8. Database Model — Milestone 2

The Milestone 1 report documented only the Attendee model. For Milestone 2 the schema is extended with Venue, Speaker and Session entities, connecting attendees, venues and speakers through sessions.



8.1 Venue Model

- id
- name
- location
- capacity
- facilities
- availability

8.2 Speaker Model

- id
- name
- email
- expertise

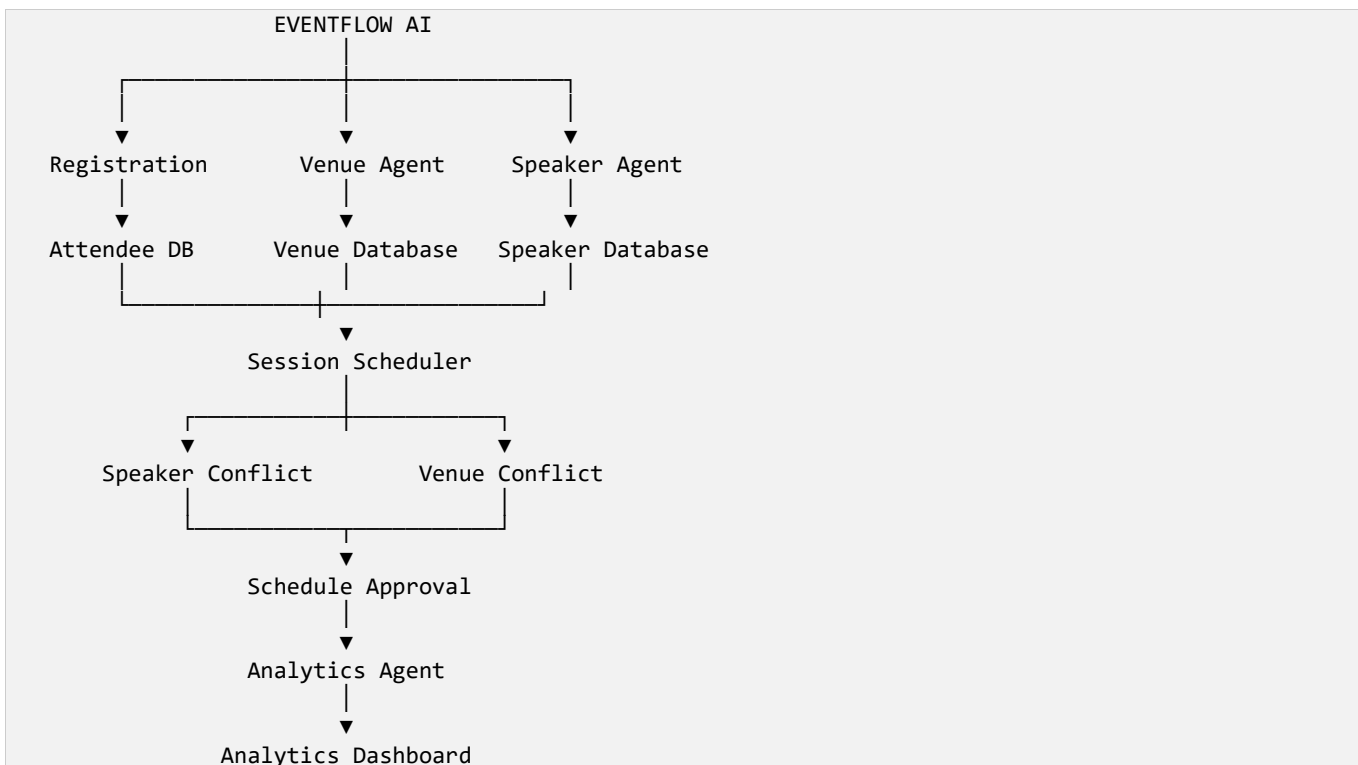
- experience
- availability
- assigned_event
- assigned_session

8.3 Session Model

- id
- title
- event
- date
- start_time
- end_time
- speaker_id
- venue_id
- status

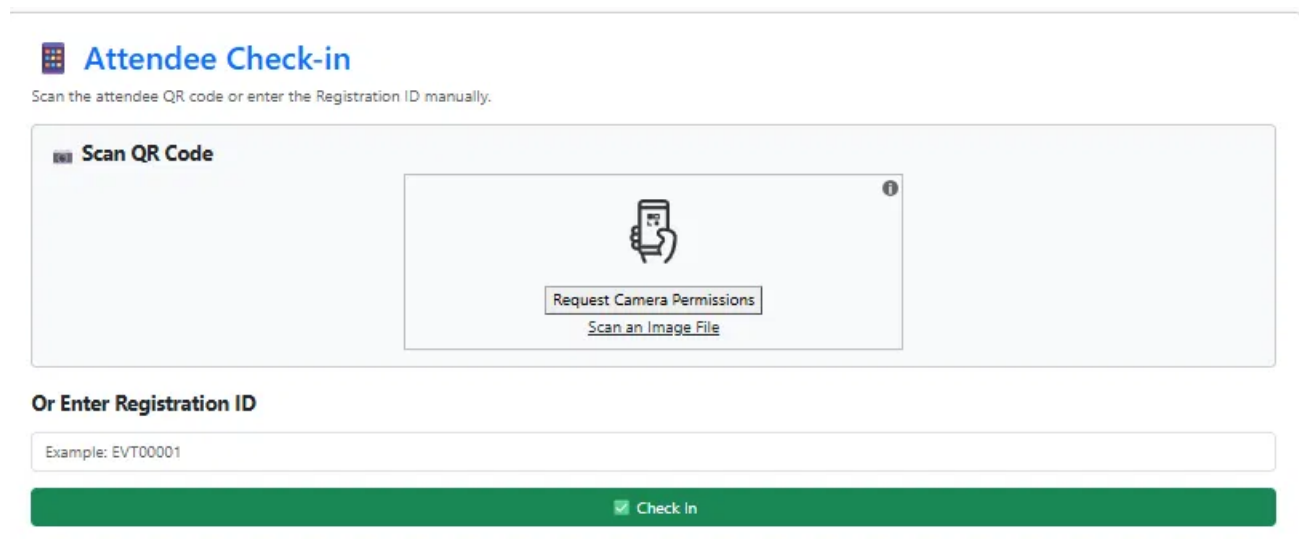
Session is the connecting entity: each session belongs to an event and references exactly one speaker and one venue, which is what allows the scheduler to detect speaker and venue conflicts across sessions.

9. Milestone 2 Architecture



10.1 QR Code-Based Check-in

EventFlow AI generates a unique QR code for each registered attendee using their registration ID. The QR code can be used during check-in to quickly identify the attendee and update their check-in status. This reduces manual entry and improves the speed and accuracy of attendee verification



Attendee Check-in

Scan the attendee QR code or enter the Registration ID manually.

Scan QR Code



Request Camera Permissions
Scan an Image File

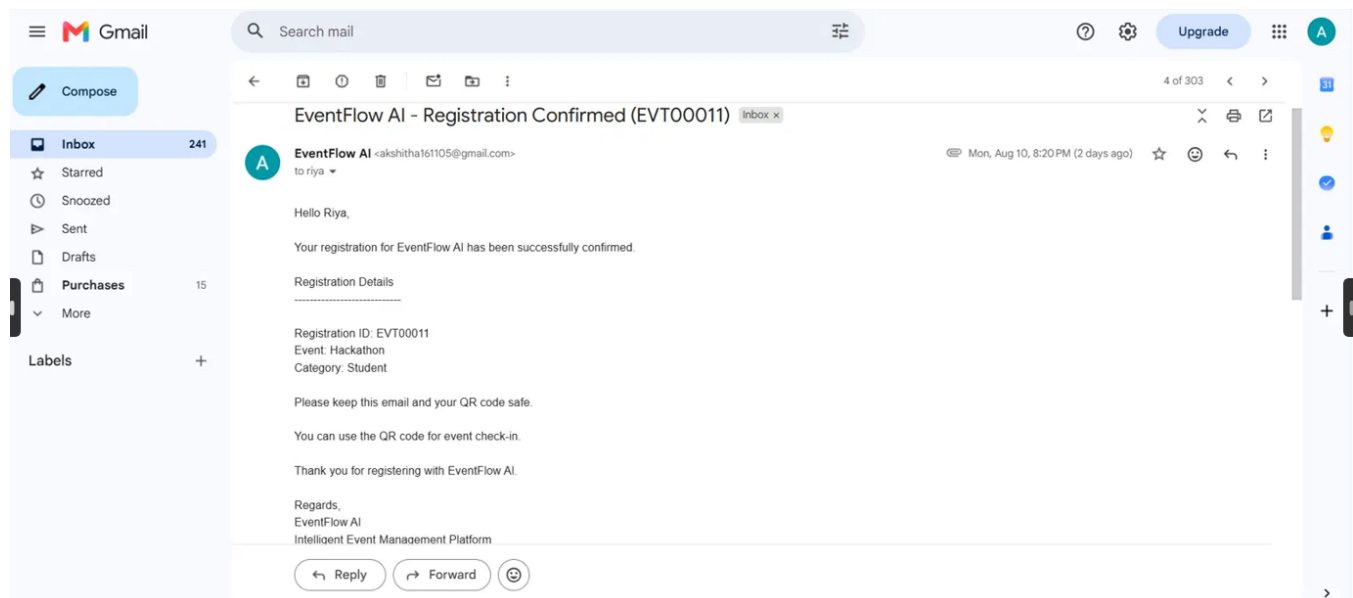
Or Enter Registration ID

Example: EVT00001

☒ Check In

10.2 Automated Email Notifications

The system provides automated email notifications for important event activities. After registration, attendees can receive a confirmation email containing their registration details and unique registration ID. Email notifications can also be used to communicate speaker session assignments and schedule updates, improving communication between organizers, attendees and speakers.



11. Implementation — Service Architecture

Milestone 2 continues the modular Flask architecture established in Milestone 1: each AI capability lives in its own service module rather than inside a single app.py.

```
services/  
  analytics_agent.py  
  email_service.py  
  registration_agent.py  
  scheduling_agent.py  
  speaker_agent.py  
  validation_service.py  
  venue_agent.py
```

12. Conclusion & Next Steps

Milestone 2 extends EventFlow AI from an attendee-registration system into an operational event-management platform. The Venue Agent recommends and ranks venues using capacity, facility and utilization scoring; the Speaker Agent manages speaker profiles and matches speakers to sessions by expertise; the Session Scheduler validates every new session against existing speaker and venue bookings before approval; and the Analytics Agent surfaces venue, speaker and session-level insights on top of the Milestone 1 attendee analytics. Together these satisfy all five Milestone 2 objectives: Venue Agent, Speaker Agent, Venue Optimization, Speaker Scheduling and Session Analytics.